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SPITFIRE IX, XI & XVI PILOTS NOTES

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{iii) *Changes of trim*

Undercarriage up ..	Nose up
Undercarriage down	Nose down
Flaps up	Nose up
Flaps down	Strongly nose down

There are marked changes of directional trim with change of power and speed. These should be countered by accurate use of the rudder trimming tab control. The firing of salvos of R/P's promotes a nosc-up change of trim; this change of trim is most marked when the weapons are fired in level flight at about 300 m.p.h. (258 kts) I.A.S.

(iv) *Flying at reduced airspeed in conditions of poor visibility.* Reduce speed to 160 m.p.h. (140 kts) I.A.S., lower the flaps and set the propeller speed control (or override) lever to give 2,650 r.p.m.; open the sliding hood. Speed may then be reduced to 140 m.p.h. (120 kts) l.A.S.

42. Stalling

(1) The stalling speeds, engine " off", in m.p.h. (knots) l.A.S. are

Aircraft without "rear-view" fuselage

	At training load (full main tanks, no ammunition or external stores) 7,150 lb.	At normal full load (full main tanks, full ammunition + 1 x 45-gallon "blister" drop tank) 7.800 lb	At maximum load (full main and rear fuselage tanks, full ammunition + 1 *x* 90-gallon "blister" drop tank) 8,700 lb.
Undercarriage and flaps up ..	90(78)	93 (80)	100 (86)
Undercarriage and. flaps down	75-79(65-69)(	80 (69)	84 (72)

Aircraft with "rear view" fuselages

Undercarriage and flaps up	95(83)	08(85)	115-117(100-102)
Undercarriage and flaps down	83-84 (71-73)	85(98)	95 (83)

The speeds above apply to aircraft which have " clipped " wings. On aircraft with " full span " wings these speeds arc reduced (at all loads by some 3-6 m.p.h. (or kts) I.A.S.

(ii) Warning of the approach of a stall is given by tail buffeting, the onset of which can be felt some 10 m.p.h. (9 kts) I.A.S. before the stall itself. At the stall either wing and the nose drop gently. Recovery is straightforward and easy.

If the control column is held back at the stall tail buffeting becomes very pronounced and the wing drop is more marked.

NOTE.—On L.F. Mk. XVI aircraft warning of the approach of a stall is not so clear; faint tail buffeting can be felt some 5 m.p.h. (or kts) l.A.S. before the stall occurs. (iii) When the rear fuselage tanks arc full there is an increasing tendency for the nose to rise as the stall is approached. This self-stalling tendency must be checked by firm forward movement of the control column, (iv) Warning of the approach of a stall in a steep turn is given by pronounced tail buffeting (and on F. Mk. XVI aircraft by hood rattling). If the acceleration is then increased the aircraft will, in general, flick out of the turn.

43. Spinning

(i) Spinning is permitted, but the loss of height involved in recovery may be very great and the following limits arc to be observed; *

(a) Spins are not to be started below 10,000 feet.

(b) Recovery must be initiated before two turns are completed.

(ii) A speed of 180 m.p.h, (156 kts) l.A.S. should be attained before starting to ease out of the resultant dive.

(iii) Spinning is not permitted when fitted with a drop tank, when carrying a bomb load, or with any fuel in the rear fuselage tank.

44. Diving

(i) At training loads the aircraft becomes increasingly tail heavy as speed is gained and should, therefore, be trimmed into the dive. The tendency to yaw to the right should be corrected by accurate use of the rudder trimming tab control.

(b) Aerobatics and combat manoeuvres are not permitted when carrying any external stores (except the 30-gallon " blister " type drop tank) nor when the rear fuselage tanks contain more than 30 gallons of fuel (but sec para. 45).

(c) When a (90 or 170) gallon drop tank or a bomb load is carried the aircraft is restricted to straight flying and only gentle manoeuvres.

(d) When wing bombs are carried in addition to a drop tank or fuselage bomb, take-off must be made only from a smooth hard runway.

(c) When carried, the 90 (or 170) gallon drop tank must be jettisoned before any dive bombing is commenced.

(/) The angle of dive when releasing a bomb or bomb load must not exceed 6o· for wing bombs of 40° for a fuselage bomb.

(g) Except in emergency the fuselage bomb or drop tank must be jettisoned before landing with wing bombs fitted. operationally.

(h) Drop tanks should not be jettisoned unless necessary operationally. While jettisoning, the aircraft should be flown straight and level at a speed not greater than 300 m.p.h. l.A.S.

(1) Except in emergency landings should not be attempted until the rear fuselage tanks contain less than 30 gallons *of* fuel. Should a landing be necessary when they contain a greater quantity of fuel the drop tank (if fitted) should be jettisoned.

53. **Position error corrections**

From To	120 150	150 170	170 210	210 240	240 290	290 350	}m.p.h. l.A.S.
Add Subtract	4	2	0 0	2	4	6	}m.p.h. or kts.
From To	106 130	130 147	147 180	180 208	208 250	250 300	}Knots l.A.S.

54. Maximum performance

(i) *Climbing*

(a) The speeds in m.p.h. (knots) for maximum rate of climb are

Sea level	to	26,000ft.	—160(140) l.A.S.
26,000ft	to	30,000ft.	—150(130) ,,
30,000ft.	to	33.000ft	.—140(122) „
33,000ft	to	37,000ft.	—130(112) „
37,000ft.	to	40,000ft.	—120(104) „
Above		40,000 ft.	—100(95)

(b) With the supercharger switch at AUTO, high gear is engaged automatically when the aircraft reaches a pre determined height (see para. 21). This is the optimum height for the gear change if full combat power is being used, but if normal climbing power (2,850 r.p.m. +12 lb./sq.in. boost) is being used the maximum rate of climb is obtained by delaying the gear change until the boost in low gear has fallen to +8 Ib./sq.in.

This is achieved by leaving the supercharger switch at MS until the boost has fallen to this figure.

(ii) *Combat*

Set the supercharger switch to AUTO and open the throttle fully.

NOTE.—On those aircraft which do not have interconnected throttle and propeller controls the propeller speed control lever must be advanced to the maximum r.p.m, position before the throttle is opened fully.

55. **Economical Flying**

(i) *Climbing*

On aircraft not fitted with interconnected throttle and propeller controls.

(a) Set the supercharger switch to MS, the propeller speed control lever to give 2,650 r.p.m, and climb at the